The University of Zambia

Department of Mathematics & Statistics

2018/2019 Academic Year, Final Examinations

MAT1110: Foundation Mathematics & Statistics for Social Sciences

Tuesday, 5 November 2019

Time Allowed: **3 hours**

Instructions:

1. There are **Seven** (7) questions in this examination paper. Attempt any five (5).
2. Indicate your **computer number** on all your answer booklets.
3. **Full credit** will only be given when **all necessary working** is shown.
4. **Calculators** are **Not** allowed.

This examination has (5) **pages** of questions.

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1. (a) i. Let $\mathbb{R}$ be the universal set, $A = (-3, 11]$ and $B = [0, 5)$. Find the set

$$A \cap B^c,$$

and display your answer on the number line.

ii. Let A and B be sets such that $A \cap B = \emptyset$. Express the following in its simplest form:

$$A - (A^c \cap B^c).$$

(b) Let $z = 6 + 4i$ and $\omega = 1 + i$ be complex numbers.

i. Express

$$\frac{z}{\omega}$$

in the form $x + iy$, where x and y are integers.

ii. Hence, or otherwise, solve the equation

$$\frac{z}{\omega} = x + ix + y + i^3 y.$$

$-3 \; 5 - 11$

$5 - \infty$

$-3 = 4$

$-3 - 0 \; 11$

$-\infty - 0$

$-\infty \; -3 - 0$

(c) Let $f(x) = \frac{7x}{3x+1}$ and $g(x) = 4 - x$.

i. On the same set of axes, sketch the graphs of $f(x)$ and $g(x)$.

ii. Find the coordinates of the points of intersection of $f(x)$ and $g(x)$.

iii. Hence, or otherwise, solve the inequality

$$\frac{7x}{3x+1} < 4 - x.$$

[8, 7, 10]

2. (a) Given that

$$f(x) = 6x^3 - 17x^2 - 4x + 3,$$

i. Show that $(2x+1)$ is is a factor of $f(x)$.

ii. Hence, or otherwise, factorize $f(x)$ completely.

(b) Solve each of the following equations;

i.

$$9^{2x+3} = \left(\frac{1}{27}\right)^{3x+1}.$$

ii.

$$\log_5 6 + \log_5 2x^2 = \log_5 48.$$

(c) On the same set of axes, sketch the graphs of

$$f(x) = 1 - \left(\frac{1}{4}\right)^x$$

and

$$g(x) = \log_{\frac{1}{4}}\left(\frac{1}{x}\right) + 1.$$

(d) Evaluate

$$\lim_{x \to 5} \frac{\sqrt{x-1} - 2}{x - 5}.$$

[6, 6, 8, 5]

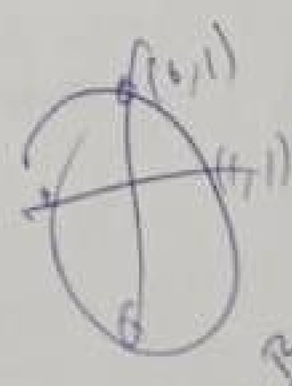
3. (a) Solve each of the following equations for $0 \le x \le 2\pi$.

i.

$$\sqrt{3}\sin x - 2\sin x \cos x = 0.$$

ii.

$$2\sin x - 1 = \csc x.$$



(b) Prove each of the following identities:

i.

$$\cos^2 x \equiv \frac{\csc x \cos x}{\tan x + \cot x}.$$

ii.

$$\frac{\csc x}{\sin x} - \frac{\cot x}{\tan x} \equiv 1.$$

(c) Given that

$$f(x) = 3\sin 2(x + 30^\circ) \text{ for } 0^\circ \leq x \leq 360^\circ,$$

i. Find the period, amplitude and phase shift.

ii. Hence, or otherwise, sketch **one revolution** (cycle) of the graph of $f(x) = 3\sin 2(x + 30^\circ)$ for $x \geq -30^\circ$.

[11, 8, 6]

4. (a) i. Use the **first principle** to differentiate

$$f(x) = \frac{1}{1 - \sqrt{x}}.$$

ii. Show that if

$$y = \frac{\ln(e^{2x^2})}{(e^{x^2})^2},$$

then

$$\frac{dy}{dx} = \frac{ax - bx^3}{e^{cx^2}},$$

where a, b and c are integers to be found.

(b) i. Integrate

$$\int \frac{(x-2)(x+3)}{\sqrt{x}}\, dx.$$

ii. Show that

$$\int_0^{\frac{\pi}{2}} e^{(\cos x + 1)} \sin x \; dx = e^2 - e.$$

(c) Find the area of the region bounded by graphs of $f(x) = x^2$ and $g(x) = 4$.

[11, 9, 5]

(d) The grouped frequency distribution below shows the masses, to the nearest gram, of 84 letters delivered by the postman.

Mass(g)	1 − 20	21 − 40	41 − 60	61 − 80	81 − 100
Number of letters	10	18	24	14	18

i. Complete the frequency distribution table below.

Mass(g)	Interval width	Frequency	Frequency density
$0.5 \le x < 20.5$	20	10	0.5

ii. Which interval represents the modal class. [7, 7, 5, 6]

7. (a) The following are test marks for a group of 120 MAT1110 students.

Mass(g)	0 − 9	10 − 19	20 − 29	30 − 49	50 − 79
Number of letters	8	21	53	28	10

i. Complete the frequency distribution table below.

Mass(g)	Interval width	Frequency	Frequency density
−0.5 − 9.5	10	8	0.8
9.5 − 19.5	10	21	2.1

ii. Draw a histogram to show the distribution of the test marks.

(b) Given that

$$f(x) = \frac{x+1}{x-1},$$

i. find $fof(x)$ leaving your answer in simplest form.

ii. find $f^{-1}(x)$.

iii. state the range of $f^{-1}(x)$.

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(c) Integrate

$$\int x^2 e^{-2x}\, dx.$$

11:30

(d) Solve the equation

$$4x^3 + 8x^2 - x - 2 = 0.$$

[8, 6, 6, 5]

5. (a) Let A and B be events with non-zero probability.

i. When are events A and B said to be Independent?

ii. State the **Bayes' theorem**.

(b) Events A and B are such that $P(A) = 0.3$, $P(B) = 0.4$ and $P(A \cap B) = 0.1$

i. Are events A and B independent? Give a reason for your answer.

Find

ii. $P(A^c \cup B^c)$.

iii. $P(A^c \cap B)$.

(c) In a survey, to investigate the effects of SHISHA smoking on University students, 30% of males and 45% of females said that they did not smoke. The Proportion of females to males was 40 : 60. If a person is chosen at random and found to be a smoker, what is the probability that this person is a male?

(d) Given that

$$f(x) = \frac{4 - 2x}{(2x+1)(x+1)(x+3)},$$

evaluate

$$\int_0^2 f(x)\, dx$$

leaving your answer in the form $\ln k$, where k is a constant.

[4, 7, 7, 7]

6. (a) Events A and B are such that $P(A|B) = \frac{2}{5}$, $P(B) = \frac{1}{4}$ and $P(A) = \frac{1}{5}$. Find

i. $P(A \cap B)$

ii. $P(B|A)$

iii. $P(A \cup B)$

(b) A group of boys at a school is entered for Advanced Level Mathematics Examinations.

Each boy takes only As examination or A2 examination or both As and A2 examinations.

The probability that a boy is taking A2 given that he is taking As is $\frac{1}{5}$.
The probability that a boy is taking As given that he is taking A2 is $\frac{1}{3}$.
Find the probability that

i. a boy selected at random is taking both As and A2 examinations.

ii. a boy selected at random is taking only As.

(c) Let

$$P(x) = x(100 - 2x),$$

where x is the number of items sold, be the profit function in dollars, for a small scale company. Find the value of x that maximizes the profit and determine the corresponding profit.